

SOLA VANUA LAVA ISLAND, Vanuatu

WMO: 915510

Lat: 13.856S

Lon: 167.539E

Elev: 59

StdP: 14.66

Time Zone: 11.00 (E11)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	70.7	72.4	63.3	87.2	76.2	64.7	91.7	76.4	22.9	77.5	20.6	77.3	1.6	20	0.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.5	89.7	81.7	88.8	81.2	88.0	80.8	82.9	87.9	82.3	87.5	81.7	86.7	3.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.5	164.1	86.5	80.8	160.2	86.0	80.1	156.8	85.3	46.9	88.5	46.1	87.7	45.5	86.9	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	15.9	12.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.3	81.6	81.5	81.4	81.3	80.4	79.5	79.0	78.7	79.4	79.6	80.6	81.2
	DBStd	2.16	1.87	1.60	1.69	1.76	1.99	2.01	1.93	2.08	2.18	2.00	1.87	1.82
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11075	980	882	974	938	942	885	898	891	881	918	919	967
	CDD65	5600	515	462	509	488	477	435	433	426	431	453	469	502
	CDH74	53669	5486	4842	5238	4887	4636	3837	3568	3430	3767	4097	4678	5203
	CDH80	12708	1794	1604	1589	1369	945	573	415	407	541	757	1167	1547
Wind	WSAvg	3.9	2.4	2.9	2.3	1.8	4.4	4.8	6.2	6.1	5.5	4.0	3.2	2.6
Precipitation	PrecAvg	160.6	15.7	13.5	16.3	17.0	14.3	10.9	10.5	9.1	10.6	13.8	15.7	13.5
	PrecMax	229.8	27.6	25.7	37.9	30.8	26.6	21.7	20.4	23.3	21.8	32.6	34.8	34.3
	PrecMin	95.9	3.3	4.8	6.8	7.2	1.9	1.8	1.6	1.4	0.7	0.4	2.2	2.1
	PrecStd	35.5	5.9	5.0	7.4	5.7	8.0	5.2	4.7	6.5	6.0	9.1	7.5	7.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.9	90.5	89.9	89.9	88.3	87.7	86.5	86.8	86.3	87.6	88.8	89.7
		MCWB	82.3	81.6	81.4	82.2	81.5	80.7	80.0	79.5	79.2	79.9	80.9	81.9
	2%	DB	89.5	89.3	88.7	88.3	86.9	85.5	84.5	84.7	85.1	86.2	87.3	88.6
		MCWB	81.6	81.2	81.3	81.4	80.4	79.3	78.2	77.8	78.3	79.1	80.0	81.2
	5%	DB	88.2	88.1	87.7	87.3	85.4	84.1	83.3	83.5	84.0	85.0	86.2	87.5
		MCWB	80.8	80.7	80.8	80.9	79.3	78.4	77.3	76.9	77.7	78.4	79.5	80.6
	10%	DB	87.0	86.9	86.5	86.1	84.0	82.8	82.1	82.3	82.8	83.8	85.1	86.2
		MCWB	80.2	80.1	80.2	80.2	78.4	77.6	76.5	76.1	77.0	77.7	78.8	80.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.6	83.5	83.2	83.7	82.5	81.7	81.0	80.7	80.7	81.2	82.4	84.3
		MCDB	89.3	89.0	88.0	88.0	87.0	86.2	85.2	86.0	84.8	86.0	87.1	88.1
	2%	WB	82.5	82.4	82.3	82.3	81.4	80.3	79.5	79.1	79.5	80.2	81.2	82.4
		MCDB	88.3	87.6	87.5	87.5	85.5	84.1	83.1	83.4	83.7	85.0	85.8	87.1
	5%	WB	81.7	81.6	81.5	81.4	80.5	79.4	78.6	78.1	78.8	79.3	80.5	81.5
		MCDB	87.2	86.7	86.6	86.3	84.4	83.0	81.8	82.1	82.9	83.6	85.0	86.2
	10%	WB	80.9	80.8	80.8	80.7	79.6	78.6	77.8	77.3	78.0	78.6	79.8	80.7
		MCDB	86.0	85.8	85.6	85.2	83.1	82.0	80.9	81.0	81.9	82.6	84.0	85.4
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.1	10.2	9.5	7.3	6.5	6.5	6.6	6.3	7.7	8.7	9.7
		MCDBR	12.7	12.9	12.3	11.5	9.2	8.4	8.2	8.7	7.8	9.9	10.5	11.9
		MCWBR	6.6	6.8	6.4	5.7	4.9	4.6	4.7	4.7	4.2	5.4	5.4	6.4
		MCDBR	12.1	11.9	11.6	11.0	9.0	8.2	7.8	8.1	7.8	9.6	10.1	11.2
	5% WB	MCWBR	6.8	6.9	6.5	6.0	5.0	4.8	5.1	4.8	4.6	5.7	5.7	6.8
Clear-Sky Solar Irradiance	taub	0.408	0.402	0.396	0.399	0.406	0.409	0.415	0.409	0.413	0.419	0.419	0.401	
	taud	2.549	2.569	2.597	2.584	2.551	2.534	2.502	2.520	2.504	2.497	2.508	2.562	
	Ebn at Noon	297	296	292	281	268	262	263	274	283	288	292	298	
	Edh at Noon	35	34	32	31	30	30	31	32	35	36	36	34	
All-Sky Solar Radiation	RadAvg	1932	1897	1745	1526	1365	1255	1321	1517	1711	1833	1891	1955	
	RadStd	194	189	167	136	129	79	75	110	137	192	153	161	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page